

2) AERIS CLI COMMAND REFERENCE

Exhaustive practical helper for day-to-day use

Keep this as your operating sheet.

AERIS CLI REFERENCE

Version

`aeris version`

Shows installed AERIS version.

Geometry commands

Show geometry status/info

`aeris geometry info`

Use this to confirm the geometry command group is available.

Generate one geometry from config

`aeris geometry generate --config configs/geometry/wing_bwb.yaml`

Short form:

`aeris geometry generate -c configs/geometry/wing_bwb.yaml`

This:

- **loads config**
- **resolves generator**
- **samples one design**
- **builds deterministic geometry**
- **writes artifacts and manifest**

Also valid for baseline config

```
aeris geometry generate --config configs/geometry/baseline_bwb.yaml
```

Dataset commands

Generate dataset with default/config sampler

```
aeris dataset generate --config configs/geometry/wing_bwb.yaml --n 10
```

Short form:

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 10
```

Generate dataset with explicit sampler

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --sampler lhs_v1
```

Generate dataset with explicit sampler seed

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --sampler lhs_v1 --seed 12345
```

```
--config configs/geometry/wing_bwb.yaml \  
--n 10 \  
--sampler lhs_v1 \  
--sampler-seed 123
```

Backward-compatible legacy LHS seed

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --lhs-seed 123
```

This still works, but conceptually it is legacy style.

Generate dataset with custom folder name

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --name my_dataset_name
```

Useful to avoid collisions.

Override plot saving behavior

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --save-plot
```

Disable:

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --no-save-plot
```

Override AeroSandbox build behavior

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --build-aerosandbox
```

Disable:

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 10 \  
  --no-build-aerosandbox
```

Full dataset command example

```
aeris dataset generate \  
  --config configs/geometry/wing_bwb.yaml \  
  --n 50 \  
  --sampler lhs_v1 \  
  --sampler-seed 777 \  
  --name wing_bwb_test_50 \  
  --save-plot \  
  --build-aerosandbox
```

Inspect an existing dataset

```
aeris dataset inspect --dataset  
data/datasets/wing_bwb_bwb_segmented_v1_n10_lhs123
```

This prints QC / summary JSON.

Pipeline commands

Run smoke pipeline

```
aeris pipeline smoke --config configs/smoke/dev.yaml
```

Short form:

```
aeris pipeline smoke -c configs/smoke/dev.yaml
```

Purpose:

- **minimal sanity workflow**
- **quick system check**

Script usage (non-CLI helper tools)

These are not the main user API, but they are useful.

Generic validation scripts***Full architecture integrity***

```
python scripts/generic/check_architecture_integrity.py
```

Checks:

- signatures
- determinism
- LHS reproducibility
- geometry pipeline
- dataset pipeline
- reproducibility across repeated runs

Metadata regeneration check

```
python scripts/generic/check_metadata_regeneration.py
```

Checks whether metadata rows match regenerated geometry.

Important:

- expects canonical dataset folder to exist

Bad config failure check

```
python scripts/generic/check_bad_config_failure.py
```

Use to verify clean failure behavior.

Large dataset sanity

```
python scripts/generic/check_large_dataset_sanity.py
```

Use for bigger dataset consistency checks.

Seed variation check

```
python scripts/generic/check_seed_variation.py
```

Checks seed-sensitive behavior.

Dataset audit

```
python scripts/generic/audit_dataset.py
```

Use for deeper dataset inspection / auditing.

BWB-specific scripts

Run one case and save geometry outputs

```
python scripts/bwb_segmented_v1/run_single_case_and_plot_3d.py \  
  --config configs/geometry/wing_bwb.yaml \  
  --output-dir data/debug/wing_bwb \  
  --no-show
```

Save planform plot too

```
python scripts/bwb_segmented_v1/run_single_case_and_plot_3d.py \  
  --save-planform
```

```
--config configs/geometry/baseline_bwb.yaml \  
--output-dir data/debug/baseline_bwb \  
--save-planform-plot \  
--no-show
```

Baseline 3D visualization

```
python scripts/bwb_segmented_v1/visualize_airplane.py --mode baseline  
--save-plot
```

This uses the script's internal baseline path.

Candidate / config visualization

```
python scripts/bwb_segmented_v1/visualize_airplane.py --mode config --  
config configs/geometry/wing_bwb.yaml --save-plot
```

If that mode exists in your current script version.

Design space inspector

```
python scripts/bwb_segmented_v1/design_space_inspector.py
```

Used to inspect/configure BWB parameter space behavior.

LHS test

```
python scripts/bwb_segmented_v1/test_lhs.py
```

Tests LHS sample generation.

LHS to geometry test

```
python scripts/bwb_segmented_v1/test_lhs_to_geometry.py
```

Tests transition from sampled design variables to realized geometry.

Utility scripts

Build dataset GIF

```
python scripts/utilities/make_dataset_gif.py
```

Used for visualization/helper output if configured.

Common output locations

Geometry run outputs

data/runs/<timestamp>_geometry_<config_name>/

Dataset outputs

data/datasets/<dataset_name>/

Debug visualization outputs

data/debug/<name>/

Examples:

- `data/debug/wing_bwb`
 - `data/debug/baseline_bwb`
 - `data/debug/visualization_runs/baseline`
-

Common operational patterns

Generate one valid geometry

```
aeris geometry generate -c configs/geometry/wing_bwb.yaml
```

Generate a reproducible dataset

```
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 100 --lhs-seed 123
```

Inspect dataset

```
aeris dataset inspect --dataset  
data/datasets/wing_bwb_bwb_segmented_v1_n10_lhs123
```

Run full integrity check

```
python scripts/generic/check_architecture_integrity.py
```

Rebuild and verify metadata consistency

```
rm -rf data/datasets/wing_bwb_bwb_segmented_v1_n10_lhs123  
aeris dataset generate -c configs/geometry/wing_bwb.yaml --n 10 --lhs-seed 123  
python scripts/generic/check_metadata_regeneration.py
```

Save baseline plot

```
python scripts/bwb_segmented_v1/run_single_case_and_plot_3d.py \
```

```
--config configs/geometry/baseline_bwb.yaml \  
--output-dir data/debug/baseline_bwb \  
--save-planform-plot \  
--no-show
```

Important practical notes

1. --lhs-seed still works

But conceptually the system is now sampler-modular.
So long-term preferred usage is:

```
--sampler lhs_v1 --sampler-seed 123
```

2. Dataset names may collide

If a dataset folder already exists, generation may fail.
Use --name to avoid collisions.

3. Baseline config must remain valid

If baseline_bwb.yaml breaks validation, plotting through config-driven path will fail.

4. Scripts are now organized by responsibility

- scripts/generic/ = architecture/system tools
- scripts/bwb_segmented_v1/ = family-specific tools
- scripts/utilities/ = helpers

That is intentional.